



Twirl (AFCO 3243)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Revision Date: 03/03/2025

Version: 1.1

SECTION 1: IDENTIFICATION

Product Identifier

Product Form: Mixture

Product Name: Twirl

Product Code: AFCO 3243

Intended Use of the Product

Use of the Substance/Mixture: Mildly alkaline powdered laundry detergent. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company

AFCO

550 Development Avenue

Chambersburg, PA 17201

T: 800-345-1329

www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

Met. Corr. 1 H290

Skin Corr. 1B H314

Eye Dam. 1 H318

STOT SE 3 H335

Aquatic Acute 2 H401

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US)



Signal Word (GHS-US)

: Danger

Hazard Statements (GHS-US)

: H290 - May be corrosive to metals
H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H335 - May cause respiratory irritation
H401 - Toxic to aquatic life

Precautionary Statements (GHS-US)

: P234 - Keep only in original container.
P260 - Do not breathe dust, mist, spray.
P264 - Wash hands, forearms, and exposed areas thoroughly after handling.
P271 - Use only outdoors or in a well-ventilated area.
P273 - Avoid release to the environment.
P280 - Wear eye protection, protective clothing, protective gloves.
P301+P330+P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P303+P361+P353 - IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+P340 - IF INHALED: Remove person to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 - Immediately call a POISON CENTER or doctor/physician.

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P312 - Call a POISON CENTER/doctor/physician if you feel unwell.
P321 - Specific treatment (see Section 4).
P363 - Wash contaminated clothing before reuse.
P390 - Absorb spillage to prevent material damage.
P403+P233 - Store in a well-ventilated place. Keep container tightly closed.
P405 - Store locked up.
P406 - Store in corrosive resistant container with a resistant inner liner.
P501 - Dispose of contents/container according to local, regional, national, territorial, provincial, and international regulations.

Other Hazards

Other Hazards Not Contributing to the Classification: Exposure may aggravate those with pre-existing eye, skin, or respiratory conditions.

Unknown Acute Toxicity (GHS-US) Not available.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances

Name	Product identifier	% (w/w)	Classification (GHS-US)
Carbonic acid, sodium salt (2:3)	(CAS No) 533-96-0	30 - 40	Skin Irrit. 2, H315 Eye Irrit. 2A, H319 STOT SE 3, H335
Sodium metasilicate	(CAS No) 6834-92-0	40 - 50	Met. Corr. 1, H290 Acute Tox. 4 (Oral), H302 Skin Corr. 1B, H314 Eye Dam. 1, H318 STOT SE 3, H335
Pentasodium triphosphate	(CAS No) 7758-29-4	10 - 20	Not classified
Octylphenol ethoxylate	(CAS No) 9036-19-5	5 - 10	Eye Dam. 1, H318 Aquatic Acute 1, H400 Aquatic Chronic 3, H412
Sodium carboxymethyl cellulose	(CAS No) 9004-32-4	0.1 - 1	Comb. Dust
Butylnaphthalenesulfonic acid, sodium salt	(CAS No) 25638-17-9	0.1 - 1	Not classified
Lemongrass 5650, OS	(CAS No) 8007-02-1	< 0.1	Flam. Liq. 4, H227
Dibutylnaphthalenesulfonic acid, sodium salt	(CAS No) 25417-20-3	< 0.1	Acute Tox. 4 (Oral), H302 Acute Tox. 4 (Inhalation: dust, mist), H332 Skin Irrit. 2, H315 Eye Dam. 1, H318 Aquatic Chronic 3, H412
Tinopal CBS	(CAS No) 27344-41-8	< 0.1	Acute Tox. 4 (Inhalation: dust, mist), H332 Eye Irrit. 2A, H319 Aquatic Acute 2, H401

The exact percentage of composition are withheld as a trade secret [29 CFR 1910.1200].

Full text of H-phrases: see section 16.

SECTION 4: FIRST AID MEASURES

Description of First Aid Measures

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

Inhalation: Remove to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER/doctor/physician if you feel unwell.

Skin Contact: Remove contaminated clothing. Drench affected area with water for several minutes. Immediately call a POISON CENTER or doctor/physician. Wash contaminated clothing before reuse.

Eye Contact: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or doctor/physician.

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Ingestion: Rinse mouth. Do NOT induce vomiting. Immediately call a POISON CENTER or doctor/physician.

Most Important Symptoms and Effects Both Acute and Delayed

General: Corrosive to eyes, respiratory system and skin.

Inhalation: Inhalation of dust may cause respiratory irritation.

Skin Contact: Causes severe skin burns. Contact may cause immediate severe irritation progressing quickly to chemical burns.

Eye Contact: Causes serious eye damage.

Ingestion: Ingestion is likely to be harmful or have adverse effects.

Chronic Symptoms: None expected under normal conditions of use.

Indication of Any Immediate Medical Attention and Special Treatment Needed

If you feel unwell, seek medical advice (show the label where possible).

SECTION 5: FIRE-FIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Use extinguishing media appropriate for surrounding fire.

Unsuitable Extinguishing Media: Do not use a heavy water stream. A heavy water stream may spread burning liquid.

Special Hazards Arising From the Substance or Mixture

Fire Hazard: Not flammable.

Explosion Hazard: Product is not explosive.

Reactivity: May be corrosive to metals. Thermal decomposition generates corrosive vapors. Avoid excessive dust production.

Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire. Under fire conditions, hazardous fumes will be present.

Firefighting Instructions: Use water spray or fog for cooling exposed containers.

Protection During Firefighting: Do not enter fire area without proper protective equipment, including respiratory protection.

Hazardous Combustion Products: Carbon oxides (CO, CO₂). Sodium oxides. Phosphorus oxides.

Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Avoid all contact with skin, eyes, or clothing. Avoid breathing dust, mist, or spray.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate unnecessary personnel.

For Emergency Personnel

Protective Equipment: Equip cleanup crew with proper protection.

Emergency Procedures: Ventilate area.

Environmental Precautions

Prevent entry to sewers and public waters.

Methods and Material for Containment and Cleaning Up

For Containment: Contain any spills with dikes or absorbents to prevent migration and entry into sewers or streams.

Methods for Cleaning Up: Clear up spills immediately and dispose of waste safely. Contact competent authorities after a spill.

Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Additional Hazards When Processed: May be corrosive to soft metals.

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do not eat, drink or smoke when using this product. Wash contaminated clothing before reuse.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Comply with applicable regulations.

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Storage Conditions: Store in a dry, cool and well-ventilated place. Keep/Store away from incompatible materials. Keep container closed when not in use.

Incompatible Materials: Strong acids. Strong oxidizers.

Special Rules on Packaging: Store in original container.

Specific End Use(s)

Mildly alkaline powdered laundry detergent with controlled foaming. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

No Occupational Exposure Limits (OELs) have been established for this product or its chemical components.

Exposure Controls

Appropriate Engineering Controls: Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. . Ensure all national/local regulations are observed.

Personal Protective Equipment: Protective goggles. Gloves.



Materials for Protective Clothing: Chemically resistant materials and fabrics.

Hand Protection: Wear chemically resistant protective gloves.

Eye Protection: Chemical goggles.

Skin and Body Protection: None usually needed.

Respiratory Protection: None usually needed.

Environmental Exposure Controls: Do not allow the product to be released into the environment.

Consumer Exposure Controls: Do not eat, drink or smoke during use.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	: Solid
Appearance	: Off-white powder
Odor	: Lemon
Odor Threshold	: Not available
pH	: 10.5 (1%)
Relative Evaporation Rate (butylacetate=1)	: Not available
Melting Point	: Not available
Freezing Point	: Not available
Boiling Point	: Not available
Flash Point	: Non-flammable
Auto-ignition Temperature	: Not available
Decomposition Temperature	: Not available
Flammability (solid, gas)	: Not flammable
Lower Flammable Limit	: Not available
Upper Flammable Limit	: Not available
Vapor Pressure	: Not available
Relative Vapor Density at 20°C	: Not available
Specific Gravity	: Not available
Solubility	: Water: Complete
Partition coefficient: n-octanol/water	: Not available
Viscosity	: Not available
Explosion Data – Sensitivity to Mechanical Impact	: Not expected to present an explosion hazard due to mechanical impact.
Explosion Data – Sensitivity to Static Discharge	: Not expected to present an explosion hazard due to static discharge.

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SECTION 10: STABILITY AND REACTIVITY

Reactivity: May be corrosive to soft metals. Avoid excessive dust production.

Chemical Stability: Stable under recommended handling and storage conditions (see section 7).

Possibility of Hazardous Reactions: Hazardous polymerization will not occur.

Conditions to Avoid: Incompatible materials.

Incompatible Materials: Strong acids. Strong oxidizers.

Hazardous Decomposition Products: Carbon oxides (CO, CO₂), Corrosive vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects - Product

Acute Toxicity: Not classified.

LD50 and LC50 Data: Not available.

Skin Corrosion/Irritation: May cause severe skin burns. (pH: 10.5 [1%])

Serious Eye Damage/Irritation: Causes serious eye damage. (pH: 10.5 [1%])

Respiratory or Skin Sensitization: Not classified.

Germ Cell Mutagenicity: Not classified.

Teratogenicity: Not classified.

Carcinogenicity: Not classified.

Specific Target Organ Toxicity (Repeated Exposure): Not classified.

Reproductive Toxicity: Not classified.

Specific Target Organ Toxicity (Single Exposure): May cause respiratory irritation.

Aspiration Hazard: Not classified.

Symptoms/Injuries After Inhalation: May cause respiratory irritation.

Symptoms/Injuries After Skin Contact: May cause burns. Contact may cause immediate severe irritation progressing quickly to chemical burns.

Symptoms/Injuries After Eye Contact: Causes serious eye damage.

Symptoms/Injuries After Ingestion: Ingestion is likely to be harmful or have adverse effects.

Chronic Symptoms: None expected under normal conditions of use.

Information on Toxicological Effects - Ingredient(s)

LD50 and LC50 Data:

Pentasodium triphosphate (7758-29-4)	
LD50 Oral Rat	3100 mg/kg
LD50 Dermal Rabbit	>7940 mg/kg
Sodium carboxymethyl cellulose (9004-32-4)	
LD50 Oral Rat	27000 mg/kg
LD50 Dermal Rabbit	>2000 mg/kg
LC50 Inhalation Rat	>5800 mg/m ³ (Exposure time: 4 h)
Tinopal CBS (27344-41-8)	
LD50 Oral Rat	>2000 mg/kg
LD50 Dermal Rat	>2000 mg/kg
LC50 Inhalation Rat	3.6 mg/l/4h
Sodium metasilicate (6834-92-0)	
LD50 Oral Rat	1153 mg/kg
Octylphenol ethoxylate (9036-19-5)	
LD50 Oral Rat	4190 mg/kg
Dibutylphthalenesulfonic acid, sodium salt (25417-20-3)	
LD50 Oral Rat	1250 mg/kg
ATE US (dust, mist)	1.50 mg/l/4h

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SECTION 12: ECOLOGICAL INFORMATION

Toxicity

Ecology - General: This material is hazardous to the aquatic environment. Keep out of sewers and waterways.

Ecology - Water: Toxic to aquatic life.

Tinopal CBS (27344-41-8)	
LC50 Fish 1	76 mg/l (Exposure time: 96 h - Species: Brachydanio rerio [static])
EC50 Daphnia 1	1000 mg/l (Exposure time: 48 h - Species: Daphnia magna)
EC50 Other Aquatic Organisms 2	10.0 - 11.0 mg/l (Exposure time: 96 h - Species: Desmodemus subspicatus)
NOEC (acute)	1.37 mg/kg (Exposure time: 14 Days - Species: Eisenia foetida [soil dry weight])
Sodium metasilicate (6834-92-0)	
LC50 Fish 1	210 mg/l (Exposure time: 96 h - Species: Brachydanio rerio [semi-static])
Octylphenol ethoxylate (9036-19-5)	
LC50 Fish 1	7.2 mg/l (Exposure time: 96 h - Species: Oncorhynchus mykiss [static])
EC50 Daphnia 1	8.6 mg/l (Exposure time: 48 h - Species: Daphnia magna [static])
ErC50 (algae)	0.21 mg/l (Exposure time: 96 h - Species: Selenastrum Green Algae)

Persistence and Degradability: Not available.

Bioaccumulative Potential

Carbonic acid, sodium salt (2:3) (533-96-0)	
BCF fish 1	(no bioaccumulation expected)
Tinopal CBS (27344-41-8)	
BCF fish 1	<1

Mobility in Soil Not available.

Other Adverse Effects

Other Information: Avoid release to the environment.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Recommendations: Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations.

Ecology - Waste Materials: Avoid release to the environment.

SECTION 14: TRANSPORT INFORMATION

14.1 In Accordance with DOT

Proper Shipping Name : CORROSIVE SOLID, BASIC, INORGANIC, N.O.S. (Disodium Trioxosilicate Pentahydrate)
Hazard Class : 8
Identification Number : UN3262
Label Codes : 8
Packing Group : III
ERG Number : 154



14.2 In Accordance with IMDG

Proper Shipping Name : CORROSIVE SOLID, BASIC, INORGANIC, N.O.S. (Disodium Trioxosilicate Pentahydrate)
Hazard Class : 8
Identification Number : UN3262
Packing Group : III
Label Codes : 8
EmS-No. (Fire) : F-A
EmS-No. (Spillage) : S-B



14.3 In Accordance with IATA

Proper Shipping Name : CORROSIVE SOLID, BASIC, INORGANIC, N.O.S. (Disodium Trioxosilicate Pentahydrate)
Packing Group : III
Identification Number : UN3262
Hazard Class : 8



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Label Codes : 8

ERG Code (IATA) : 8L

14.4 In Accordance with TDG

Proper Shipping Name : CORROSIVE SOLID, BASIC, INORGANIC, N.O.S. (Disodium Trioxosilicate Pentahydrate)

Packing Group : III

Hazard Class : 8

Identification Number : UN3262

Label Codes : 8



SECTION 15: REGULATORY INFORMATION

US Federal Regulations

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SARA Section 311/312 Hazard Classes	Immediate (acute) health hazard.
Carbonic acid, sodium salt (2:3) (533-96-0)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Pentasodium triphosphate (7758-29-4)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Sodium carboxymethyl cellulose (9004-32-4)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Water (7732-18-5)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Tinopal CBS (27344-41-8)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Sodium metasilicate (6834-92-0)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Octylphenol ethoxylate (9036-19-5)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Dibutylphenol sulfonic acid, sodium salt (25417-20-3)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Butylphenol sulfonic acid, sodium salt (25638-17-9)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	

US State Regulations

Pentasodium triphosphate (7758-29-4)	
U.S. - New Jersey - Discharge Prevention - List of Hazardous Substances	
U.S. - New York - Reporting of Releases Part 597 - List of Hazardous Substances	
RTK - U.S. - Pennsylvania - RTK (Right to Know) - Environmental Hazard List	
RTK - U.S. - Pennsylvania - RTK (Right to Know) List	
U.S. - Texas - Effects Screening Levels - Long Term	
U.S. - Texas - Effects Screening Levels - Short Term	
Sodium carboxymethyl cellulose (9004-32-4)	
U.S. - Texas - Effects Screening Levels - Long Term	
U.S. - Texas - Effects Screening Levels - Short Term	
Sodium metasilicate (6834-92-0)	
U.S. - Texas - Effects Screening Levels - Long Term	
U.S. - Texas - Effects Screening Levels - Short Term	
Octylphenol ethoxylate (9036-19-5)	
U.S. - Texas - Effects Screening Levels - Long Term	
U.S. - Texas - Effects Screening Levels - Short Term	
Tinopal CBS (27344-41-8)	
U.S. - Texas - Effects Screening Levels - Long Term	

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U.S. - Texas - Effects Screening Levels - Short Term

Canadian Regulations

Water (7732-18-5)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Tinopal CBS (27344-41-8)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Carbonic acid, sodium salt (2:3) (533-96-0)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Pentasodium triphosphate (7758-29-4)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Sodium carboxymethyl cellulose (9004-32-4)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Sodium metasilicate (6834-92-0)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Octylphenol ethoxylate (9036-19-5)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Dibutylphthalenesulfonic acid, sodium salt (25417-20-3)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Butylphthalenesulfonic acid, sodium salt (25638-17-9)

Listed on the Canadian DSL (Domestic Substances List) inventory.

Lemongrass 5650, OS (8007-02-1)

Not listed on the Domestic Substances List (DSL) or the Non-Domestic Substances List (NDSL) inventory.

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR.

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision date : 03/03/2025

Other Information : This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

GHS Full Text Phrases:

Acute Tox. 4 (Inhalation: dust, mist)	Acute toxicity (inhalation: dust, mist) Category 4
Acute Tox. 4 (Oral)	Acute toxicity (oral) Category 4
Aquatic Acute 1	Hazardous to the aquatic environment - Acute Hazard Category 1
Aquatic Acute 2	Hazardous to the aquatic environment - Acute Hazard Category 2
Aquatic Chronic 3	Hazardous to the aquatic environment - Chronic Hazard Category 3
Comb. Dust	Combustible Dust
Eye Dam. 1	Serious eye damage/eye irritation Category 1
Eye Irrit. 2A	Serious eye damage/eye irritation Category 2A
Flam. Liq. 4	Flammable liquids Category 4
Met. Corr. 1	Corrosive to metals Category 1
Skin Corr. 1B	Skin corrosion/irritation Category 1B
Skin Irrit. 2	Skin corrosion/irritation Category 2
STOT SE 3	Specific target organ toxicity (single exposure) Category 3
H227	Combustible liquid
H232	May form combustible dust concentrations in air
H290	May be corrosive to metals
H302	Harmful if swallowed
H314	Causes severe skin burns and eye damage

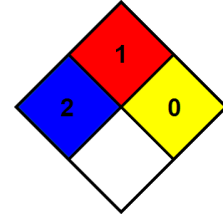
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H315	Causes skin irritation
H318	Causes serious eye damage
H319	Causes serious eye irritation
H332	Harmful if inhaled
H335	May cause respiratory irritation
H400	Very toxic to aquatic life
H401	Toxic to aquatic life
H412	Harmful to aquatic life with long lasting effects

- NFPA Health Hazard** : 2 - Intense or continued exposure could cause temporary incapacitation or possible residual injury unless prompt medical attention is given.
- NFPA Fire Hazard** : 1 - Must be preheated before ignition can occur.
- NFPA Reactivity** : 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.



HMIS III Rating

- Health** : 2 Moderate Hazard - Temporary or minor injury may occur.
- Flammability** : 1 - Slight Hazard.
- Physical** : 0 - Minimal Hazard.

Party Responsible for the Preparation of This Document

AFCO
550 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

North America GHS SDS 2015 (U.S., Can., Mex.)